

A comprehensive review on the development of data-driven methods for wind power prediction and AGC performance evaluation in wind–thermal bundled power systems

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ABSTRACT

The wind–thermal bundled power system achieves energy complementarity and optimized scheduling, which is an important way to build a new type of energy system. For the safe and stable operation of the wind–thermal bundled power system, accurate data-driven analysis is necessary to maintain real-time balance between electricity supply and demand. By summarizing the development and characteristics of wind–thermal bundled power system in China and different countries, current research in this field can be clearly defined in two aspects: short-term wind power prediction for wind farms and performance evaluation of automatic generation control (AGC) for thermal power generation units. For short-term wind power prediction, it is recommended to focus on historical data preprocessing and artificial intelligence methods. The technical characteristics of different data-driven wind power prediction methods have been compared in detail. For performance evaluation of AGC units, a comprehensive analysis was conducted on current evaluation methods, including the “permitted-band” and “regulation mileage” methods, as well as the issue of evaluation failure in traditional evaluation methods in practical engineering. Finally, the relative optimal dynamic performance of AGC units was discussed and the future trend of data-driven research in wind–thermal bundled power system was summarized.

1. Introduction

Wind–thermal bundled power system refers to a power generation structure in which wind and thermal power resources work together in the same area, complement and coordinate with each other to achieve large-scale energy transmission and efficient operation of the power system. In this type of power system, wind power, as a renewable energy source, provides “green electricity” in a low-carbon way [1]. However, wind power is greatly affected by climate conditions and its output is fluctuating. As a traditional energy source, thermal power provides large-scale stable output and sufficient flexibility, which can be used as a frequency regulation resource to address the challenges brought by wind power. The main role of wind–thermal bundled power system is to achieve complementarity between wind and thermal power, improve the overall operational efficiency of the power system, reduce the

impact on the environment, and achieve the transformation and upgrading of energy structure. To achieve this goal, it is necessary to pay attention to the output fluctuations under high penetration of wind power, improve the frequency regulation resource capacity of thermal power units, and optimize the scheduling strategy of regional power grids [2].

In the next decade, the development speed of wind power generation in the world will triple to maintain net zero emissions and reduce the negative impacts of climate change [3]. In terms of wind power market share, it is dominated by China, followed by the United States, the United Kingdom, Brazil, and Vietnam [4]. Taking China as an example, in 2023, the proportion of renewable energy generation to the total electricity consumption in society has reached 15.3 %. It is expected that by 2060, the proportion of renewable energy generation in China will exceed 60 %, and the energy system needs to be transformed into a more

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flexible new form. On the side of the power supply, renewable energy is mainly used; On one side of the power grid, it is necessary to enhance the ability of peak shaving, frequency regulation, and voltage regulation; On the user's side, more diverse distributed energy storage is needed.

Authoritative research institutions such as the International Energy Agency (IEA), International Renewable Energy Agency (IRENA), Energy Information Administration (EIA), and British Petroleum Corporation (BP p.l.c.) have conducted extensive research on the low-carbon transformation of energy systems, providing useful references for energy transformation research [5–9]. However, the research methods proposed by these organizations are generally limited to the analysis of the total supply and demand of various types of energy, and insufficient consideration is given to the impact of the internal supply and demand balance of the energy system, especially the power system.

In the context of high penetration rate of renewable energy and immature long-term energy storage technology, the randomness and volatility of renewable energy pose challenges to the balance of electricity supply and demand [10]. In power grids of different countries and regions, there are high requirements for power quality. Controlling the frequency fluctuation of the power grid within a reasonable range around 50 Hz or 60 Hz is a key guarantee for the safety of the power grid [11]. Therefore, it is necessary to continuously monitor the frequency changes of the power grid in real-time, by adjusting controllable equipment to maintain a balance between the supply and demand of active power. China is currently the world's largest wind power country [12]. According to statistics from the National Energy Administration (NEA), as of the end of April 2023, the installed capacity of wind power in China was 380 million kilowatts, accounting for 14.3 % of the country's installed power generation capacity [13]. At the same time, China is also the world's largest country in thermal power, accounting for 50 % of the total global thermal power capacity [14]. Thermal power dominates the basic form and characteristics of China's power system, playing a crucial role in ensuring reliable power supply and supporting system frequency, and is difficult to effectively replace in the short term.

The practical engineering experience of energy transmission shows that fully utilizing the adjustability of thermal power units helps solve the problem of wind power fluctuations and improve the utilization of energy transmission channels. In addition, technological development in the fields of ultra-high voltage transmission [15], flexible direct current transmission [16,17], and integrated pipeline corridors [18] has provided a good foundation for multi-source power transmission between regions, and interconnectivity also helps to play a mutually beneficial role among regional power grids. Therefore, the intensive development of renewable energy and the wind–thermal bundled power system have become one of the important measures to build a new type of energy system, and are a reasonable way of energy development and utilization [19].

2. Development and technical challenges of wind–thermal bundled power system

2.1. The engineering development of wind–thermal bundled power system

There are many practical projects related to wind–thermal bundled power system in different countries. The Roscoe wind farm within the jurisdiction of Texas in the United States was once one of the largest onshore wind farms in the world [20]. In actual operation, this wind farm and other types of power generation facilities such as thermal power plants achieve complementary collaboration through the scheduling system of the Electric Reliability Council of Texas (ERCOT) to meet changes in electricity demand. A typical wind–thermal bundled project in Germany is the collaborative operation of the DolWin3 offshore wind power group and nearby thermal power plants. The DolWin3 offshore wind power group is located in the North Sea of Germany, with a total installed capacity of 900 megawatts [21]. Although the DolWin3 project mainly focuses on wind power transmission, it connects two wind farms

in the southwestern North Sea of Germany to the onshore power grid. The DolWin3 project, along with thermal facilities and other forms of energy, participates in the operation of the German electricity market. The Cofrentes nuclear power plant operated by Iberdrola in Spain provides important basic electricity supply and is coordinated with nearby wind and solar power plants for scheduling [22]. The installed capacity of Cofrentes is 1092 megawatts, achieving efficient operation of the power system through complementary scheduling with surrounding renewable energy power plants [23]. For the Hallett wind farm and the nearby Torres Island thermal power plant operated by the same energy company AGL Energy in Australia, the total installed capacity of the Hallett wind farm is 350 megawatts. The Torres Island thermal power station is a gas turbine power station with an installed capacity of 1280 megawatts. The distance between these two projects is about 200 km, and the efficient operation of the energy system has been achieved through the scheduling of the wind–thermal bundled power system [24].

As the world's largest country in wind power and thermal power, China has many typical cases of wind–thermal bundled power system. The three northern regions of China (including the northeast, north, and northwest of China) are rich in wind energy resources. The Qinghai–Henan Ultra High Voltage Direct Current (UHVDC) transmission project is the world's first power export project mainly focused on large-scale renewable energy transmission [25]. The sending end of the project, namely the Qinghai power grid, is connected to a large amount of photovoltaic and wind power, and through AC/DC conversion technology, the electricity is sent to Henan Province. Based on the Zhangbei ± 500 kV High Voltage Direct Current Transmission based on Voltage Source Converters (VSC-HVDC) project and the Zhangbei–Xiong'an 1000 kV Ultra-High Voltage Alternating Current (UHVAC) transmission project, hundreds of wind farms and thousands of photovoltaic power stations in Zhangjiakou and Chengde regions of Hebei Province have been combined to form a multi-source complementary energy system, solving the problems of unreasonable power supply structure and insufficient peak shaving capacity of the power grid [26]. In the Xilingol League in central Inner Mongolia, leveraging its abundant wind energy resources, the 500 kV collection station of the million-kilowatt-grade wind power base project of Huaneng North Shangdu was put into operation in September 2022 [27]. In addition, wind power from the western region of Inner Mongolia is also transmitted to the North China region through the ultra-high voltage transmission channel (Mengxi–Tianjin South UHVAC transmission project), playing a key role in ensuring the power consumption of the Beijing–Tianjin–Hebei load center [28]. In March 2023, the construction of China's first "Wind–Solar–Thermal-Storage integration" large-scale comprehensive energy base external transmission project, namely Longdong–Shandong ± 800 kV UHVDC transmission project, began construction in the Northwest Power Grid [29]. The project meets the transmission requirements of thermal and renewable energy in the Gansu Energy Base and contributes to the reliable operation of the Northwest Power Grid.

Through the various typical projects related to wind–thermal bundled power system mentioned above, it can be seen that the strategy of coordinated operation of wind power and thermal power (as well as other types of energy) is of great significance in modern energy systems and is widely applied globally. From a technical perspective, wind–thermal bundled power system balances the volatility and uncertainty of renewable energy, while utilizing the stability of thermal power generation to ensure the stable and reliable operation of the energy system. The complementary scheduling strategy helps to improve the scheduling flexibility and operational efficiency of the power grid. Secondly, these practical projects also demonstrate that the practical applications of wind–thermal bundled power system in different geographical environments and operating markets have good diversity and applicability.

2.2. The correlation and technical challenges between wind power forecasting and AGC performance evaluation

Frequency regulation of the power grid by thermal power units is achieved through the Automatic Generation Control (AGC) system. The structure of a typical provincial wind–thermal bundled power system is shown in Fig. 1. Driven by the goal of energy conservation and emission reduction, the proportion of renewable energy connected to the power system will continue to increase in the future. As shown in Fig. 2, generally, the scale of wind power in wind–thermal bundled power system is relatively large, and the output of wind power is intermittent and fluctuating. When there is an imbalance between the supply and demand of active power in the grid, according to the frequency regulation principle of the power system, the frequency of the grid fluctuates. Energy storage technology benefits from millisecond level response, it has significant advantages in improving power system stability and reliability. However, due to the impact of energy storage in terms of technological maturity, capacity scale, and investment return, there are still many challenges in solving large-scale wind power consumption through energy storage [30]. Therefore, for the frequency regulation of regional power grids, especially secondary frequency regulation, thermal power units are still the main force at present.

By monitoring the load fluctuations in the energy system and the output of each thermal power generation unit, the active power of the unit is controlled in real-time to maintain the frequency stability of the power system. The random fluctuations of wind power usually occur within a relatively shorter time scale, while thermal power units often demonstrate delays and deviations when completing corresponding regulation, and the time scale of the control process is relatively long. In actual operation, thermal power units are influenced by factors such as factory design, service life, operating conditions, and control systems in actual operation, and there are individual differences among different thermal power units. The power grid dispatch department must conduct refined and personalized analysis of the performance of the thermal power units participating in frequency regulation, in order to adjust control strategies in real-time and issue more accurate and reasonable control instructions.

Wind power, as a clean and low-carbon renewable energy source, is greatly affected by climate conditions, and the intermittency and random fluctuations are the main characteristics of wind power. Intermittency refers to the fact that wind power is dependent on wind speed and direction, and the lack of wind leads to a decrease in power generation. Therefore, wind power output is not continuously stable, but intermittent. Random fluctuation refers to the unpredictability of wind power. Due to the difficulty in accurately predicting changes in wind intensity and direction, the output of wind power has randomness at different time scales. These two characteristics pose challenges to the stable operation of the energy system, as the power grid needs to maintain a balance between power supply and demand at all times. On the other hand, as a traditional energy source, thermal power has a

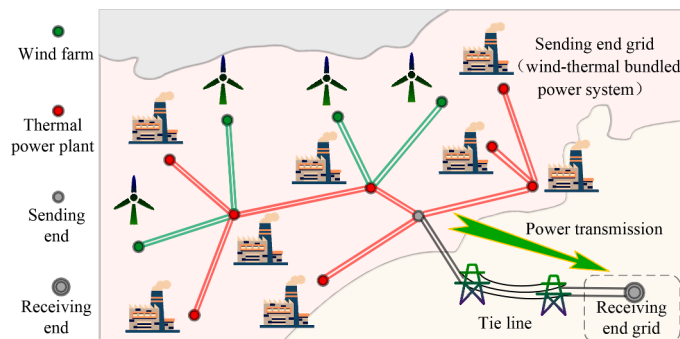


Fig. 1. Typical structure of a provincial wind–thermal bundled power system in northern China.

series of excellent characteristics that make it play a crucial role in the energy system. The biggest advantage of thermal power lies in its strong schedulability and stable output, which is crucial for the operation of the energy system.

The intrinsic connection between wind power prediction and the performance evaluation of thermal AGC units is mainly reflected in the following aspects:

- Requirement for Frequency Regulation and Peak Shaving:** The intermittency and randomness of wind power induce fluctuations in the power grid frequency. Accurate wind power forecasting aids grid operators in preparing sufficient frequency regulation and peak shaving resources in advance. The performance evaluation of thermal power AGC units, as a significant method for frequency regulation and peak shaving, is directly related to their capability to counterbalance the variability in wind power.
- Operating Costs and Efficiency:** The accuracy of wind power prediction affects the economic dispatch of thermal power AGC units. Inaccurate predictions may lead to frequent AGC interventions, increasing fuel consumption and operational costs, thereby impacting the overall efficiency of the power grid.
- Renewable Energy Integration:** Enhancing the precision of wind power prediction enables a greater utilization of wind energy resources and reduces reliance on thermal power generation.

In summary, in the context of thermal power units serving as the main force of frequency regulation, in order to achieve large-scale wind power consumption, there are mainly two technical challenges in the research of wind–thermal bundled power generation systems:

(1) Enhancement of short-term wind power forecasting performance

In wind–thermal bundled power system, the output of thermal power units is stable and can be controlled according to the actual requirements of the power grid. However, the regulation ability of wind turbines is relatively weak, or due to economic considerations, these functions of wind turbines have not been frequently used [31]. When the grid's frequency regulation and voltage regulation capabilities are insufficient, the strategy of cutting off wind turbines or even shutting down entire wind farms is often adopted. This not only leads to wind curtailment, but also easily causes fatigue damage to the mechanical components of wind turbines, and even affects their service life [32]. Wind curtailment leads to incalculable economic losses and weakens the competitiveness of wind power in the energy market [33]. Therefore, improving the accuracy of wind power prediction is of great significance for energy balance and power dispatch [34], and helps to solve the problem of renewable energy consumption [35].

(2) Accurate evaluation of the AGC performance of thermal power units

The dynamic performance evaluation of automatic generation control (AGC) for thermal power units reveals their characteristics under various operating conditions. The performance evaluation of AGC distinguishes the performance differences between different units in different power plants, which helps to provide optimization basis for scheduling strategies. At the same time, the performance evaluation of AGC also helps to provide a reasonable reference for technical transformation of thermal power plants, reduce operating costs, and improve fuel utilization efficiency. Secondly, the evaluation results also help the power grid dispatch center to monitor the actual contribution of thermal power units in frequency regulation, ensure the fairness and impartiality of economic assessment, and promote the sustainable development of auxiliary service frequency regulation market.

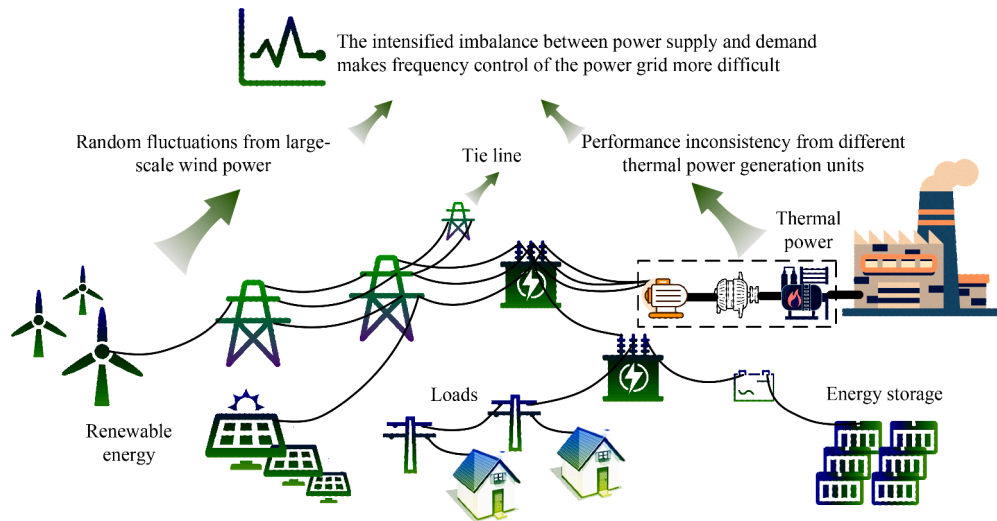


Fig. 2. Main technical challenges of wind-thermal bundled power system in modern energy systems.

3. Development of data-driven wind power prediction methods

According to the Global Wind Energy Council (GWEC)'s 2023 Global Wind Report, as of 2022, the total installed capacity of wind power worldwide has reached 906 GW [36]. It is expected that by 2024, the new installed capacity of global onshore wind power will exceed the milestone of 100 GW for the first time, and the new grid connected capacity of global wind power will reach 680 GW in the next five years. Therefore, the intermittency and random fluctuations of large-scale wind power are currently one of the main challenges faced by the energy system.

The wind power active power curve of a provincial power grid in China within 30 days is shown in Fig. 3. Due to the intermittency and random fluctuations of wind power, as well as the complexity and unpredictability of meteorological information, it is very difficult to control wind power. Based on current research, improving the accuracy of wind power prediction mainly relies on the following two technical means:

3.1. Numerical simulation and meteorological information perception methods

In the process of wind power prediction, numerical simulation and meteorological information perception methods, as traditional prediction methods, have shown significant advantages in accuracy, but they face some challenges in practical applications. For example, based on the analysis of real-time meteorological data [37], a wind process simulation calculation model covering wind farm terrain roughness, air pressure changes and fan wake effect is constructed [38], and then the wind speed and direction over a period of time are predicted by solving fluid dynamics and thermodynamic equations [39]. Furthermore,

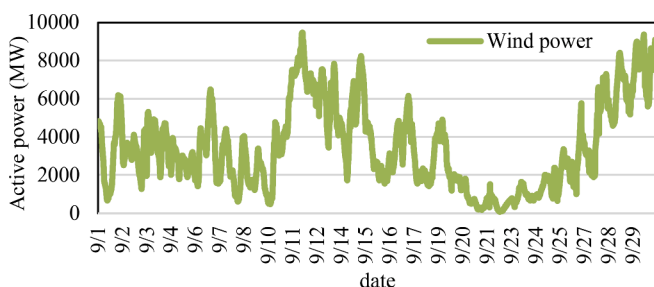


Fig. 3. Wind power curve of a provincial power grid in China within 30 days.

Doppler laser radars were used to obtain the wind information from a certain distance to improve the wind power prediction [40]. For this technical step, the methods require massive geographic and meteorological information. Therefore, they are easily affected by the accuracy of the instruments and the objective environment, and the application of these methods is difficult and costly. In addition, in the actual operation of wind farms, considering the limitations of maintenance costs imposed by operators, the dimensions of data collected on site by most wind turbines are relatively limited, usually only including wind direction and speed.

3.2. Historical data processing and artificial intelligence methods

In the daily operation of wind farms, a large amount of wind power data and meteorological data are accumulated, and the distribution of relevant data samples is relatively chaotic [41]. By reasonable data cleaning, outliers containing bad data and noise data can be removed to increase the proportion of effective samples and provide high-quality training sets for prediction models [42]. At present, traditional data cleaning methods mainly rely on the mapping between wind speed and wind power, while ignoring the correlation between other wind condition variables. Therefore, research on data cleaning methods in the multi-dimensional data space of wind power is still blank. In addition, considering the intermittency and random fluctuations of wind processes, complex regularities are often hidden among different variables in wind farm data [43]. Efficient extraction of these hidden features is the key to improving the accuracy of wind power prediction. At the same time, the development of graphics processing unit (GPU) acceleration technology has greatly improved the training efficiency of deep neural networks [44]. Therefore, a data-driven and algorithmic optimization based neural network prediction model is currently one of the most economical and effective technical means to improve the accuracy of wind power prediction. Relevant research in this area mainly focuses on the following two aspects: wind power data preprocessing and data-driven wind power prediction models.

(1) Wind power data preprocessing

For actual wind farms, even in seasons with relatively stable wind conditions, a considerable portion of bad data and noise data may appear. These abnormal data are often caused by equipment failures, sensor failures, weather changes, and other factors, and the proportion of abnormal data in historical data is usually relatively low. However,

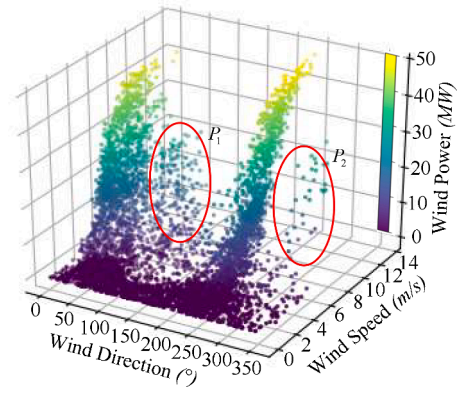
abnormal data may lead to deviations in the training process of wind power prediction models. For example, when the power value corresponding to abnormal data is an abnormal state, these erroneous results may cause the model to learn the inaccurate relationship between wind power and wind speed, thereby affecting the accuracy of prediction results. Therefore, it is necessary to preprocess the historical data of wind farms in a reasonable manner [45].

There are many conventional methods for wind power data preprocessing, such as filling in missing data through interpolation, using standardization to eliminate dimensional effects [46], using wavelet transform for time-domain decomposition [47], and using clustering to analyze the correlation between predicted values and historical data [48]. Reference [49] proposes an optimal intra group variance algorithm to identify outliers. The samples composed of wind speed and wind power are located in a two-dimensional plane. Based on the numerical value of wind speed, all samples can be divided into several adjacent intervals, and the variance of wind power within each interval is obtained. For a single sample, once the deviation between wind power and the average value exceeds a certain threshold, the sample can be identified as an outlier within this wind speed interval. Although variance is sensitive to outlier detection, if there are multiple outliers in the dataset, it may lead to an increase and deviation in the overall variance. Moreover, due to the influence of multiple meteorological factors on wind power, identifying outliers based solely on a single wind speed may be slightly biased. In reference [50], an equivalent boundary model for detecting outliers is proposed. In the two-dimensional mapping of wind speed wind power in this model, the rotational speed of the wind turbine is incorporated to form a three-dimensional joint boundary. The proposed boundary takes into account the uncertainty of variables and forms an elastic operating boundary based on a scaling coefficient for identifying abnormal data.

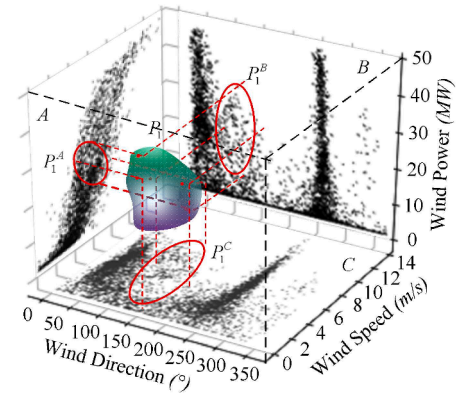
The three-dimensional sample space of wind direction, wind speed and wind power from a wind farm in China within 60 days is shown in Fig. 4. As shown by the red circles in Fig. 4(a), there are two sparse regions in the sample space, namely P_1 and P_2 . The densities of P_1 and P_2 in the sample space are significantly different from those dense samples, hence P_1 and P_2 are often defined as outliers. The projections of P_1 and P_2 are shown in Fig. 4(b). As can be seen from Fig. 4, the output of wind turbines not only changes with wind speed, but also with other weather conditions, especially wind direction. Due to changes in wind direction, the blades of wind turbines are subjected to different airflow directions and intensities. In order to analyze the output of wind farms more accurately, it is necessary to conduct multi-dimensional collaborative analysis of different meteorological variables, which helps to accurately detect outliers in historical datasets and improve the accuracy of wind power prediction.

The sample distribution shown in Fig. 4 essentially represents the power curve of wind turbines. It is evident that the distribution of wind power may have different characteristics under varying wind speed conditions. Reference [52] notes that this distribution is referred to as an asymmetric and heterogeneous conditional distribution. Consequently, the literature proposes a probabilistic wind power curve modeling method based on conditional variational autoencoders, employing an Infinite Gaussian Mixture Model (IGMM) to better model this heterogeneous conditional distribution. Under the framework of conditional variational autoencoders, the established wind power curve is clearer and more reliable. The experimental results within the wind turbine Supervisory Control and Data Acquisition (SCADA) dataset indicate that the proposed method exhibits enhanced robustness in the preprocessing of multivariate input data. Similarly, reference [53] indicates that incorporating operational variables such as blade pitch angle and rotational speed into the Gaussian regression model can significantly reduce the uncertainty in the predicted power curve.

(2) Data-driven wind power prediction model



(a) Sample distribution



(b) Outlier projection

Fig. 4. A three-dimensional sample space of wind direction, wind speed and wind power from an actual wind farm within 60 days [51].

From the perspective of modeling methods, wind power prediction models can be divided into two categories: machine learning/statistical models, and deep learning models; from the perspective of prediction results, wind power prediction can be divided into deterministic prediction, and interval prediction (also known as probabilistic prediction). Among them, deterministic prediction is to provide specific wind power values for a certain future time or period, while interval prediction or probabilistic prediction is to provide corresponding uncertainty in the prediction results, that is, to represent the upper and lower limits of the prediction results through confidence intervals or prediction intervals. Based on these two classification perspectives, the technical characteristics of different data-driven wind power prediction methods were sorted and compared, and the results are shown in Table 1.

When a large number of wind power samples are gathered in multi-dimensional space, the distribution of the samples reflects the regularity of the historical dataset. The data-driven deep learning model is effective at capturing the complex relationships in wind power data. In recent years, more and more deep learning models have been applied to wind power prediction. Including Convolutional Neural Networks (CNN) [62], Long Short-Term Memory Networks (LSTM) [65], Bidirectional LSTM (BiLSTM) [66], Gated Recurrent Units (GRU) [67], Bidirectional GRU (BiGRU) [68], Deep Confidence Networks (DBN) [69], and Autoencoders (AE) [70], these technologies are receiving increasing attention in wind power prediction due to their outstanding ability to solve complex nonlinear problems.

The typical Convolutional Neural Network (CNN) was proposed by LeCu in 1989 [71]. With the support of GPU, CNN has demonstrated strong feature extraction capabilities, gradually becoming an ideal choice for processing large-scale and complex datasets. In reference

Table 1

Comparison of technical characteristics of different data-driven wind power prediction methods.

	Machine learning/statistical models Core methods	Strengths and weaknesses	Deep learning models Core methods	Strengths and weaknesses
Deterministic prediction	Autoregressive Integrated Moving Average (ARIMA) [54]: A model in statistics used for analyzing and predicting time series data. Mainly including: autoregressive model (AR), integral model (I), and moving average model (MA).	·High interpretability and reliability. ·The prediction of non-periodic and non-seasonal data is not ideal. ·It is necessary to perform stationarity testing and differential processing on temporal data, and statistical experience is required.	Stacked Denoised Autoencoder (SDAE) [55]: A deep learning model that stacks multiple Autoencoders together, improving the generalization ability of the model by adding multiple hidden layers and denoising.	·Used for automatic extraction of wind power features. ·Compared to ordinary Autoencoders, it shows advantages in noise reduction. ·The internal structure is complex, making it difficult to explain the decision-making and feature extraction process of the model.
	Decision Tree (DT) [56]: A classification and regression method based on tree structure for handling nonlinear relationships and interactive features. Compared to linear regression, it exhibits stronger robustness.	·Easy to explain, and able to handle missing or abnormal values. ·If noise data or abnormal data affect the splitting point of the decision tree, the performance of the model will be affected. ·Need to cooperate with pruning and other regularizations to avoid overfitting.	Multi-level Transfer Learning (MTL) [55]: Divide the knowledge similarity between the source domain and the target domain into different levels, and select different data transfer strategies based on the level correlation. For example, using source wind field data as initialization parameters or auxiliary training data for pre trained models.	·Compared to ordinary transfer learning, multi-level structures are beneficial for improving learning performance. ·Beneficial to improving the power prediction accuracy of newly built, expanded or rebuilt wind farms. ·There may be a negative transfer effect, where the source domain interferes with the target domain.
	XGBoost [57]: Based on DT, an ensemble learning algorithm commonly used for processing high-dimensional, sparse data. Showing good explanatory for the mapping between meteorological variables and wind power, commonly used to provide deterministic prediction results [58].	·Based on special regularization, effectively prevents overfitting of the prediction model. ·Demonstrated the ability to automatically process missing values without filling or deleting data. ·Difficulty in learning the deep representation of wind data.	Transformer [59]: A deep learning model based on attention mechanism, which is good at capturing key information in sequences and exhibits good parallel computing ability for processing time series data.	·Feature reconstruction of datasets through trend enhancement and periodic augmentation. ·The visual model decision-making is provided by the attention matrix, demonstrating good interpretability. ·The temporal order relationship between the elements of a sequence may not be explicitly captured.
	Reinforcement Learning (RL) [60]: An algorithm that learns how to make optimal decisions in a scenario through interaction with the environment. The Q-learning algorithm was used in this literature to generate optimal ensemble weights to combine three basic learners and improve wind power prediction performance.	·By generating optimal ensemble weights, combining the advantages of different learners to improve wind power prediction. ·Continuously update the Q table through environmental interaction to achieve real-time optimization. ·Due to the influence of initial state and reward function, careful design and parameter adjustment are required.	Graph Convolution Network (GCN) [61]: A deep learning model based on graph structure, used to process spatially irregular and time-dependent data, which can effectively extract and represent features of irregular graph structure data.	·Suitable for scenarios with irregular spatial distribution, such as wind farms. ·Proficient in extracting temporal and spatial correlations between different nodes, improving prediction accuracy. ·Used to solve multi-site and multi-step prediction problems in wind farms.
Interval prediction	Quantile Regression (QR) [62]: A regression analysis method developed from traditional conditional mean regression, which is more robust than traditional linear regression.	·Based on the quantiles of samples rather than the mean, the prediction results are not sensitive to outliers. ·A large amount of data is needed to estimate the different quantiles of the target variable, otherwise there may be overfitting issues.	Autoencoder (AE) [58]: The basic structure is an encoder and decoder, where the encoder compresses the input data into a low dimensional encoding, while the decoder decompresses the encoding back to the original data. Used for wind data preprocessing.	·High dimensional input data is compressed into low dimensional encoding, reducing model complexity and improving model inference speed. ·Easy to show overfitting. ·Poor interpretability of extracted wind power features.
	Gradient Boosting (GB) [63]: Belongs to ensemble learning methods, which combine a series of weak prediction models (usually DT) to establish a strong prediction model and automatically select important features.	·The impact of specific features on prediction results can be measured, and the interpretability of the model is better. ·Early stages of training are susceptible to noise data. ·The requirement for data quality is high, and the model requires data preprocessing and feature engineering.	Lower Upper Bound Estimation (LUBE) [58]: A deep learning model based on upper and lower bounds estimation. By maximizing the coverage of the prediction interval and minimizing the normalized average width of the prediction interval, the boundaries of probabilistic predictions are generated by LSTM.	·Providing confidence intervals for deterministic prediction models (such as LSTM), providing more comprehensive prediction results. ·The training and inference time of the model is relatively long, which may affect real-time performance.
	Variational Mode Decomposition (VMD) [64]: A signal decomposition method that decomposes a signal into multiple Empirical Mode Decomposition (EMD), each EMD corresponds to different frequencies and amplitudes. Usually used for data preprocessing.	·Good interpretability of signal characteristics. ·High computational complexity, requiring multiple iterations. ·The ability to process large-scale wind power data is limited, and the decomposition results are influenced by initial parameters.	Temporal Convolutional Networks (TCN) [64]: Compared to CNN, dilation convolution is adopted in TCN, which expands the receptive field without increasing the number of parameters, thereby better capturing long-term dependencies in the sequence.	·Residual blocks and attention mechanisms are introduced to improve temporal prediction performance. ·The dilation convolution operation may lead to a sharp increase in model complexity and memory consumption, and the model has limitations on the length of the input sequence.

[72], a sequence to sequence (Seq2Seq) prediction model based on CNN is proposed. In this model, the encoder and decoder are the two main components of the network. The encoder is used to extract features from the input time series, and the decoder is used to convert abstract features into target sequences. However, the Seq2Seq model is generally suitable for situations where the output length of the model is uncertain, such as machine translation, speech recognition, and text generation tasks. For wind power prediction, the length of the output result is fixed, so a linear connection layer should be used instead of the Decoder to obtain the required prediction results and reduce the complexity of the model.

Reference [73] proposes a CNN based wind speed prediction model suitable for one-dimensional data, and divides wind direction and wind speed into multiple independent inputs, which are fed into the model as training samples. However, if the feature variables are independent of each other when input into the model, then each time the model extracts features, it only captures the features of a single meteorological variable. For wind power prediction, there is a complex interaction between wind direction, wind speed, and wind power, and their characteristics must be integrated together to maximize the advantages of CNN. Reference [74] proposed a method to convert one-dimensional wind power data into two-dimensional data, and achieved the fusion of wind direction and wind speed data. However, when combining these inputs, it did not pay attention to the impact of specific arrangement and combination methods on the prediction results. Although this method of simply arranging the data in order and then inputting it into CNN achieved relatively good results. But in fact, this effect benefits from the powerful feature extraction ability of CNN. If the arrangement of input features is combined with the principle of local receptive field in CNN, so that CNN can capture the correlation characteristics between multiple meteorological variables every time it scans the feature map, it will inevitably improve the performance of the prediction model. Because this is equivalent to providing customized input samples for CNN, achieving the reconstruction of input features. Wind power prediction is a typical time series prediction problem that requires consideration of features in the time dimension. At the same time, spatial features should also be considered, including the spatial correlation between fans and the spatial correlation between inputs. Reference [75] proposed a spatio-temporal feature fusion method that extracts spatial and temporal correlations between input variables through graph convolutional neural networks (GCN) and multi resolution convolutional neural networks (MCNN). Thus, the spatial correlation of the extracted features is visualized, and the interpretability of the model is improved.

In the study of wind power prediction based on deep learning, to extract temporal features, the input to the model is typically organized as multivariate time series data. It is noteworthy that Recurrent Neural Networks (RNNs) [76] have been demonstrated to perform exceptionally well in time series prediction. LSTM, a variant of RNN, addresses the vanishing gradient problem through a gating mechanism and is capable of handling long sequences and long-term dependencies, thus avoiding the short-term memory issue. LSTM exhibits long-term dependent learning ability and the advantage of avoiding gradient vanishing compared to traditional RNN, but requires more training parameters, more memory usage, and longer training time. GRU can be seen as a variant of LSTM, which couples the forgetting gate with the input gate. Compared to LSTM, GRU outputs two values, requires fewer training parameters, less memory, and exhibits faster training speed. However, when dealing with long sequences, the prediction accuracy of the model is not ideal. In summary, LSTM is suitable for processing large sequences or datasets that require high accuracy, while GRU is suitable for situations that require less memory and faster running speed.

Reference [77] points out that in a single deep learning model, CNN performs best in predicting wind speed/wind power, thanks to its powerful feature extraction ability. Reference [51] used data cleaning and feature reconstruction with CNN, and compared the proposed method with some classic methods such as LSTM, ElmanNN, ARIMA kal. Moreover, this study used different error evaluation indicators for wind

power prediction, including mean absolute error (MAE), root mean square error (RMSE), mean absolute percentage error (MAPE), and symmetric mean absolute percentage error (SMAPE). The data cleaning (DC) method proposed in this study increased the proportion of available data in the historical dataset of wind farms and improved the data quality of training samples; The proposed feature reconstruction (FR) further highlights the short-term and ultra short-term temporal features in the input data. The short-term wind power prediction results and corresponding errors under different prediction methods are shown in Fig. 5. Based on a large amount of error analysis, this study concluded that compared to classic deep learning models, CNN based prediction models exhibit the best performance, which is consistent with the conclusion in reference [77].

In recent years, research on RNNs and their variants has achieved numerous exciting breakthroughs, and the combination of RNNs with self-attention mechanisms has enabled the analysis of ultra-long time series data. Reference [78] adopts a bilateral branch learning framework that integrates high and low sampling resolution data to enhance the accuracy of wind power prediction. The essence of this method is the employment of CNN and LSTM, along with Multi-Head Self-Attention Mechanisms (MHSA), to model and optimize the prediction of wind power. Reference [79] innovatively combines attention mechanisms with Ordinary Differential Equation Networks (ODEN) and LSTM to address the issue of quantile crossing in probabilistic wind power forecasting. This network architecture helps the model to focus on the most crucial segments of the time series, while the advantages of ODEN are leveraged to break away from the conventional usage of discrete data. Reference [80] proposed a two-channel deep network model based on multi-source data to enhance the accuracy of ultra-short-term wind power prediction. The model combines geographic information of wind farms with multivariate time series data and employs a GCN-LSTM structure to extract spatio-temporal individual dynamic features, along with classical feature selection methods for feature engineering, thereby enhancing prediction accuracy. The study introduces a graph matrix-based method to predict the power output of newly installed wind turbines without the need for SCADA data from these turbines, potentially aiding in the expansion and upgrading of wind farms. In the realm of recurrent encoder-decoder networks, Reference [81] proposes the Stochastic Recurrent Encoder-Decoder Neural Network (SREDNN). SREDNN extends the stochastic recurrent model to accommodate exogenous variables, such as meteorological conditions, within the encoder-decoder network framework for wind power prediction.

In data-driven renewable energy forecasting tasks, data from multiple renewable energy power stations are involved, which may contain sensitive information such as the location and output of the power stations. If such data were leaked, it could pose a threat to the security and privacy of the power stations. Therefore, in the latest research, scholars have focused on the data security issues of wind farms. Reference [82] proposes a novel bi-party engaged data-driven modeling framework (BEDMF) that achieves data privacy protection by isolating data on local clients, and by capturing the spatiotemporal characteristics among multiple sites, features are input into data-driven models as decentralized data.

In summary, current research in wind power forecasting has made significant progress, particularly in terms of model accuracy and data processing. However, future efforts need to focus more on achieving breakthroughs in ensuring the generality and adaptability of predictive models to accommodate constantly changing environmental conditions and wind farm layouts. At the same time, as the importance of data privacy protection becomes increasingly prominent, research must also further strengthen the exploration of data security in wind power, develop more innovative data protection technologies, and ensure data security and privacy protection in the field of renewable energy.

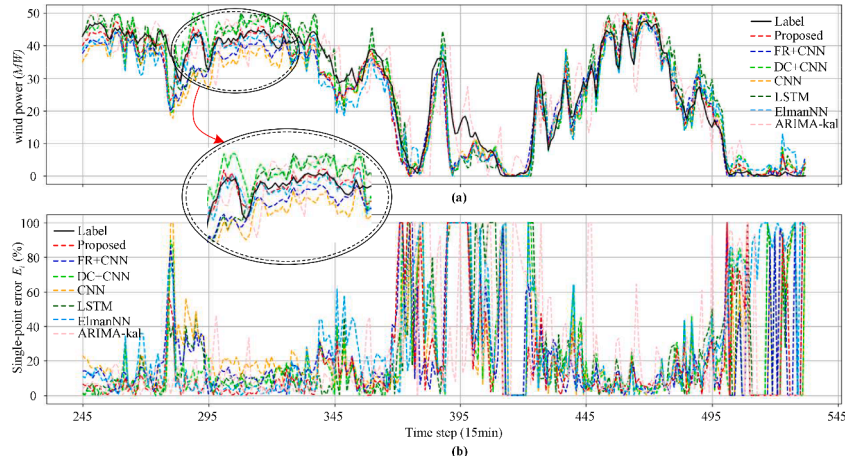


Fig. 5. Short term wind power prediction results and corresponding errors under different prediction methods (a) Prediction and actual output of wind farms using different methods (b) Single point error curves of different methods [51].

4. Development of data-driven AGC performance evaluation methods for thermal power units

In wind-thermal bundled power system, wind power prediction provides advanced scheduling decisions for thermal AGC units in the power grid, and AGC units ensure the balance of active power supply and demand by adjusting their own output in real-time. However, the AGC system is a typical multi-input multi-output complex nonlinear system, which is influenced by engineering factors such as unit design, service time, operating conditions, and control systems in practical projects. Therefore, the performance of different thermal power AGC units is significantly inconsistent. When controlling the output of the AGC unit, it is necessary to accurately grasp the unit's response ability to AGC commands, and consider a series of indicators such as the unit's adjustment speed and accuracy. This enables the AGC unit to respond quickly and accurately to AGC commands, thereby stabilizing the frequency of the power grid. Therefore, as the proportion of renewable energy continues to increase, accurate evaluation of the performance of thermal AGC units is crucial for the energy system.

In actual thermal power plants, the generation of boiler steam requires multiple combinations of wind, coal, and water, and requires multiple energy forms of conversion to complete it. Starting from receiving the command from the coordinated control system, it often takes 3–5 min for the thermal power unit to adjust the fuel quantity, and then to experience a change in steam production [83]. This process is

influenced by multiple equipment, parameters, and operating conditions. From the perspective of automatic control, the control object of thermal power AGC technology is the generator set. A typical thermal power generation system mainly consists of two parts, namely the coal-burning subsystem and the water-steam subsystem, as shown in Fig. 6. It can be seen that the thermal power generation system involves many operating parameters, and the actual working conditions are extremely complex.

The control target of AGC is to meet the real-time balance of power supply and demand under the premise of achieving high-quality power supply [85]. A schematic diagram of the control target of AGC is shown in Fig. 7. Different AGC units receive different power set-point (P_{SET}) from the dispatch center, and each AGC unit adjusts its own active power output (P_{OUT}) to track P_{SET} as closely as possible. Both P_{SET} and P_{OUT} are sequential, where P_{SET} is a dynamically changing step signal and P_{OUT} is a nonlinear signal [84].

In the operation of thermal power units, the matching between the main steam pressure and load must be precise to ensure the efficiency of frequency regulation in the current power grid, it is necessary to analyze the performance differences of AGC units in detail and provide accurate evaluation results. The research and application of AGC mainly includes the following two aspects:

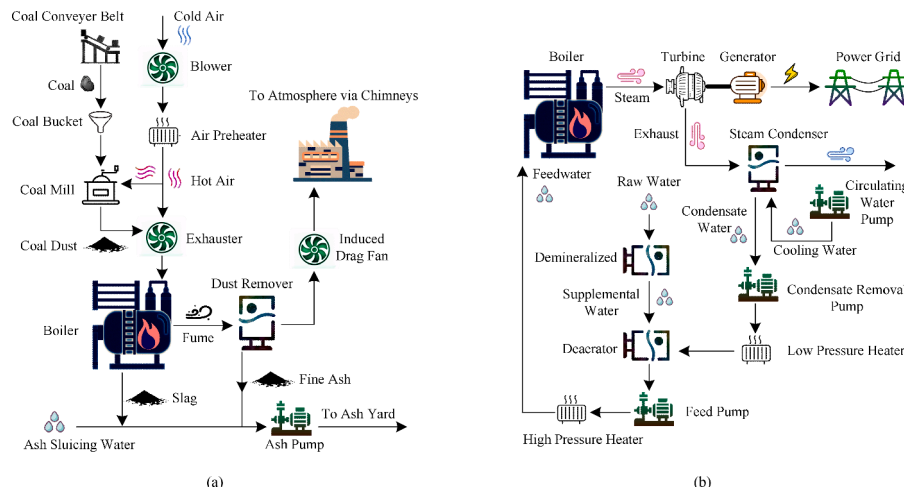


Fig. 6. A typical operation process of thermal power generation: (a) The coal-burning subsystem, (b) The water-steam subsystem [84].

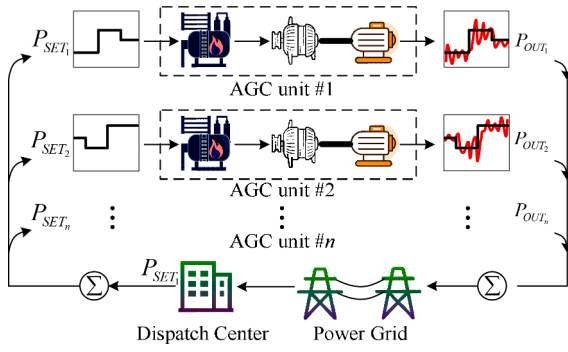


Fig. 7. The schematic diagram of the control target of AGC [84].

4.1. Frequency control at the regional power grid level

The research in this field belongs to model driven, and power grid operators mainly focus on the overall control strategy and control effectiveness of regional power grids, including the rationality and economy of scheduling, as well as the stability of power quality indicators such as grid frequency and transmission power of tie lines.

In order to achieve optimal coordinated control of the overall scheduling of regional power grids, a virtual generation ecosystem control strategy based on the idea of time tunnel is proposed by reference [87]. This strategy introduces a two-layer dynamic power allocation structure, combines hierarchical learning with consensus theory to improve the adaptability of consistency algorithms in complex random environments, and achieves autonomy and optimal control of microgrids. An optimal AGC control method for a two-zone interconnected power system is proposed by reference [88]. The hybrid PID fuzzy controller proposed in this study improves the frequency stability of the interconnected two zone power system and exhibits better dynamic performance than traditional PID controllers. The optimal parameters of the controller are determined based on classical intelligent algorithms. Reference [89] proposed a meta heuristic algorithm to find the optimal solution by simulating the hunting behavior of eagles, and applied it to frequency and voltage regulation of high renewable energy permeability power systems. The algorithm was validated based on modeling of capacitive energy storage and virtual inertial units. The results indicate that by storing short-term energy on the DC side of distributed power electronic devices (such as converters or inverters) and injecting it into AC measurement, the frequency stability of the system can be improved.

The above research on AGC mainly focused on the control of the overall performance of the power grid, which essentially belongs to online economic dispatch (ED). These studies did not focus on the difficulty of tasks undertaken by different AGC units in daily operation, as well as the output differences of AGC units in daily operation.

4.2. Dynamic performance evaluation at the level of thermal AGC units

The research in this field is data-driven, focusing on the response ability and characteristics of AGC units. With the influence of factors such as coal quality and equipment aging, the dynamic performance of thermal AGC units may change due to their load-bearing capacity and service duration [90]. Although optimizing the parameters of each unit controller may improve the consistency of AGC performance to some extent, different thermal power units still exhibit different output characteristics even under the same AGC command [91].

A model prediction based AGC control method was proposed by reference [92], which improves the frequency response margin of the power system. The proposed method considers offline and online identification of power system parameters as a whole. Using offline recognition to estimate the probability distribution of key parameters in historical operating data, and then monitoring and updating the

probability distribution of these parameters in real-time through online recognition. This study points out that using deep learning models, such as LSTM, to predict future loads and renewable energy generation power is beneficial for accurately adjusting the output of AGC units. A data-driven calculation method for the required capacity of AGC frequency modulation was proposed by reference [93]. The proposed method utilizes conditional probability to establish a correlation between frequency modulation capacity, active power fluctuation, and frequency modulation performance. And interval prediction of the standard deviation of future loads was carried out using an extreme learning machine algorithm. The research approach of establishing connections between different types of AGC historical data and utilizing data mining techniques has demonstrated good reference value for accurately revealing the characteristics of AGC units. The study of reference [94] began to focus on the dynamic performance of AGC units, especially thermal power units. The proposed study indicates that the dynamic performance of AGC units (with relatively shorter time scales) will constrain ED (with relatively longer time scales). This is because traditional ED processes typically run every 5 min to provide basic scheduling points and adjust backup capacity, but do not consider the impact of AGC dynamic performance. The dynamic AGC data collection cycle is generally 2–6 s, in order to update data in real-time, calculate Area Control Error (ACE), monitor grid performance, and maintain frequency stability of the power system. Therefore, it is necessary to accurately evaluate the dynamic performance of AGC units in order to achieve system flexibility management. Especially for power systems with high renewable energy penetration rates, the rationality of evaluation indicators is crucial.

In summary, the above typical studies have focused on the three core technical aspects of AGC units:

- (a) Model prediction and parameter identification. By combining offline and online system parameter identification, as well as using deep learning models for load and renewable energy generation prediction, the accuracy of AGC control has been improved.
- (b) Data driven demand calculation. By constructing the correlation between frequency modulation capacity, active power fluctuation, and frequency modulation performance, based on machine learning algorithms, the operational characteristics of AGC units are finely analyzed.
- (c) Accurate evaluation of dynamic performance. The dynamic performance of AGC units, especially thermal AGC units, is of great significance for economic dispatch. Therefore, in power systems with high renewable energy penetration rates, it is necessary to accurately evaluate the real-time dynamic performance of AGC units.

Among the methods proposed in the above literature, the advantages of model prediction and parameter identification methods lie in real-time tracking and updating of system parameters, improving prediction accuracy. However, the challenge lies in the investment of a large amount of computing resources and the requirement for high-quality data. The advantage of data-driven demand calculation method lies in fully utilizing historical data and revealing the operational characteristics of AGC units. However, the challenge lies in the possibility that the performance of the method may be affected by the amount of data, and it also requires relevant professional knowledge for data processing and analysis. The research advantage of dynamic performance evaluation lies in its ability to improve the power quality of regional power grids in environments with high penetration of renewable energy, but it may require more data and complex models to obtain refined evaluation results. Overall, the above related research still faces a series of challenges such as data quality, computational resources, and model complexity.

By organizing the methods for evaluating the performance of AGC

units in different regions, they can be further classified into two categories: one is the “permitted-band” method used in Europe (such as Spain and Germany), and the other is the “regulation mileage” method, represented by the PJM/MSIO electricity market in the United States and the SGCC/CSG electricity market in China.

4.2.1. The “permitted-band” evaluation method

The performance evaluation method based on the “permitted-band” in Spain and Germany is shown in Fig. 8. The solid red line is P_{OUT} . The solid black line is P_{SET} , which represents the required ideal performance of the unit, so it is also called the fastest permitted response. Correspondingly, the dashed green line is the slowest permitted response [84].

For the power grid in Spain, the slowest permitted response nearly equal to a first-order exponential curve with a time constant of 100 s. In the German power grid, an initial response delay of 30 s is permitted after the unit has received a new P_{SET} step, and each P_{SET} step must be fully completed within 300 s. So the slowest permitted response in Germany is a straight line with a slope equal to the step value of the current P_{SET} divided by 270 s [95].

In Fig. 8, the light green area between the dashed green line and the P_{SET} is the permitted response band. In addition, the dashed yellow line is the error tolerance boundary, which is generally a percentage of P_{SET} or a certain value (e.g. $\pm 5\%P_{SET}$ in Europe, ± 2 MW in China) [96]. The light yellow area between the dashed yellow line (error tolerance boundary), the dashed green line (slowest permitted response) and the P_{SET} is the error tolerance band.

The behavior of the unit's P_{OUT} exceeding the error tolerance band is unexpected and will be subject to financial penalties, corresponding to the purple area in Fig. 8. For some special units that habitually have an offset between P_{SET} and P_{OUT} , strictly limiting the “permitted-band” may ignore the actual contribution of the unit to the power grid. For example, when the actual output of certain units shows overshoot or undershoot relative to AGC instructions, the output of the units is not strictly operating within the “permitted-band”. However, the overall change of the unit output has achieved the requirement of P_{SET} , which is still beneficial to the entire power grid. Hence the other method called “regulation mileage” overcomes this evaluation problem.

4.2.2. The “regulation mileage” evaluation method

The performance evaluation method based on the “regulation mileage” is shown in Fig. 9, which is also the current standard in the United States and China electricity markets. For each continuous AGC set-point power, the “regulation mileage” referring to the absolute value of the difference between the output of the AGC unit at the beginning and the end of each P_{SET} . As illustrated in Fig. 9 left, in American power grid, each pair of black and red arrows represents the required regulation mileage and the actual regulation mileage, respectively [31]. Here it is assumed that the movement towards the P_{SET} is positive, movement away from the P_{SET} is negative. It grants full remuneration if the actual regulation mileage is more than 70 % of the required regulation mileage during 5-minute dispatch interval [97]. Under this evaluation system, a multi-agent based deep reinforcement learning algorithm framework was proposed by reference [98] to achieve coordinated load frequency

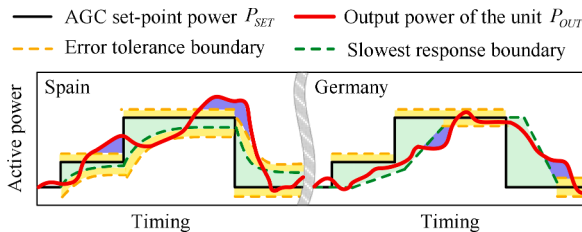


Fig. 8. The principle of “permitted-band” AGC dynamic performance evaluation in the Spanish and German electricity markets [84].

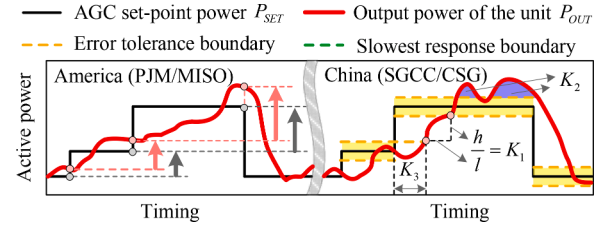


Fig. 9. The principle of “regulation mileage” AGC dynamic performance evaluation in the US and Chinese electricity markets [84].

control in multi region integrated energy systems. In order to achieve the optimal performance and comprehensive economic benefits of AGC in each region, this study analyzed the calculation of the adjustment mileage cost and pointed out that the adjustment mileage cost is a part of the AGC objective function used to evaluate the economic benefits of AGC. In addition, the study also provided a calculation method for adjusting the mileage cost, including the calculation formulas for adjusting the mileage and adjusting the mileage cost, but no related improvements and optimizations have been proposed for the performance evaluation method.

The Chinese power market further refined the three dynamic performance indicators of the i -th AGC unit in the j -th regulation, that is, regulation speed K_1 , regulation accuracy K_2 , and response delay K_3 [99]. As shown in the schematic diagram of China (SGCC/CSG) in Fig. 9.

The regulation speed K_1 refers to when the unit receives a new AGC set-point power and starts to increase output, starting from its reliable crossing of the current error tolerance boundary until it enters the new error tolerance boundary for the first time. Record the duration of this range as l , and the difference in output at the starting point as h , then the regulation speed $K_1 = h/l$. The regulation speed refers to the speed at which the unit responds to commands, which can be divided into an increase rate and a decrease rate.

The regulation accuracy K_2 refers to the difference between the actual output of the unit and the AGC set-point power within the range from the first time the unit enters the new error tolerance boundary to the end of the current AGC set-point power, represented by a purple area. Integrate the absolute value of the difference between the actual output P_{OUT} of the unit and the set-point power P_{SET} during this period, and then divide the integration value by the integration time to obtain the calculated value of K_2 .

The response delay K_3 refers to the time it takes to reliably cross the error tolerance boundary consistent with the adjustment direction for the first time since receiving a new AGC set-point power. In practice, it is also necessary to standardize K_1 , K_2 , and K_3 to eliminate dimensional effects.

In the K -method evaluation system, the value of K is related to the adjustment depth and output of the AGC unit, and is one of the important factors that affect compensation benefits. It can be seen that among AGC units of the same type, the daily regulation mileage of AGC units is determined by the number of daily calls, and the compensation income of AGC units is determined by the daily regulation mileage and comprehensive performance K -value. At present, there are few relevant studies that have improved and optimized the K -value evaluation system for thermal AGC units.

The three indicators K_1 , K_2 , and K_3 meticulously depict the dynamic performance of AGC units. However, based on the above analysis, the calculation process of performance indicators is very sensitive to whether the output of the unit can successfully cross the “error tolerance boundary”. Therefore, although the AGC performance evaluation system in the Chinese power market is currently the strictest, there is still a problem of performance evaluation failure in some special scenarios.

4.3. Optimal dynamic performance evaluation of AGC and corresponding challenges

In the energy system, AGC is mainly provided by conventional power sources such as hydroelectric, gas, and thermal power units. Due to the fact that these power sources are composed of mechanical devices with rotational inertia, converting primary energy into electrical energy undergoes a series of complex processes. Especially for thermal AGC units, there is a deviation between their frequency regulation performance and the regulation expectations of the power grid, specifically the delay and deviation (overshoot and undershoot) of regulation. In addition, since the AGC instruction is a dynamically changing step signal, its duration and step amplitude are also dynamically changing. If the historical data of the unit is not fully utilized, it is difficult to detect whether each AGC unit has reached its own "optimal state", that is, the relative optimal dynamic performance.

Studies have proposed improvements on the optimal dynamic performance evaluation of thermal power AGC units based on conventional methods. In reference [100], filtering and feature extraction were performed on the unit output P_{OUT} in the time domain and frequency domain, respectively. Separating the primary response from the secondary response highlights the trend of the unit's response to AGC commands. Reference [101] points out that due to the high nonlinearity and complexity of AGC systems, the response delay of different units also varies, and the impact of response delay must be considered in power market. Therefore, a discrete time model for thermal power generation units has been proposed, which consists of nonlinear modules and linear modules. The nonlinear module has the functions of error tolerance bandwidth limitation and power change rate limitation, while the linear module consists of second-order linear submodules and delay offset submodules. This model effectively reproduces the actual output characteristics of AGC units, but for large-scale provincial power grids, accurately modeling each AGC unit and restoring its parameters is a very cumbersome task, and when the unit performance changes again with the increase of service time, it is necessary to re-model. In fact, the power grid dispatch center stores a large amount of historical operating data of AGC units, and through data mining technology, the regular characteristics of the units can be fully described and characterized. In addition, the study also pointed out that the response delay of the unit is not only affected by the performance of key equipment, but also by the resolution of AGC instructions.

Some generalized dynamic performance models and evaluation indicators for AGC units have been proposed by relevant research. The time series error between unit output and AGC instructions is defined in reference [102], and then scaled according to different electricity market rules. The concepts of Service Performance Index (SPI) and Quality of Service (QoS) are proposed. The performance indicators proposed in this study distinguished whether the response of AGC units was qualified, and were validated using short-term (approximately 30 s) primary frequency modulation response data. Reference [103] used and validated real operating data from AGC units. This study divided the nonlinear time series output curve of the unit into several adjacent short line segments, and compared the endpoint change amplitude of each segment with the change amplitude of AGC instructions to obtain an evaluation of dynamic performance. This method is based on the assumption that the AGC unit maintains a relatively stable upward or downward trend for a certain period of time, and the sampling window of the analyzed data should not be less than 5 min, otherwise it is difficult to distinguish any trend characteristics of the unit output. That is to say, this method analyzes AGC instructions with a time scale longer than 5 min. Reference [84] proposed a dynamic performance error judgment method based on energy regulation, which constructs a method composed of AGC instruction duration T_{AGC} (min), instruction amplitude ΔP_{AGC} (MW) and dynamic performance error $\bar{E}_{SDI}$. The AGC sample space was segmented and a data-driven method was proposed to

determine the optimal operating point in several truncated intervals, as shown in Fig. 10.

In recent years, numerous studies have made valuable contributions to the performance evaluation of AGC units. Reference [104] introduces a method that employs max-relevance and min-redundancy (mRMR) for feature extraction from historical parameter series related to AGC regulation. Furthermore, it utilizes an LSTM network for the prediction of AGC regulatory capacity. By computing the AGC regulation index and designating it as the target function for optimization, the study provides a reliable evaluative basis for thermal power AGC units. Reference [105], through simulation modeling based on Dymola, conducted an in-depth study of the dynamic characteristics of 300 MW units in thermal power plants, verifying the effectiveness of the model during steady-state processes and peak shaving operations. Moreover, the study assessed the economic performance of the units under different load change rates in conjunction with the economic indicators of the power

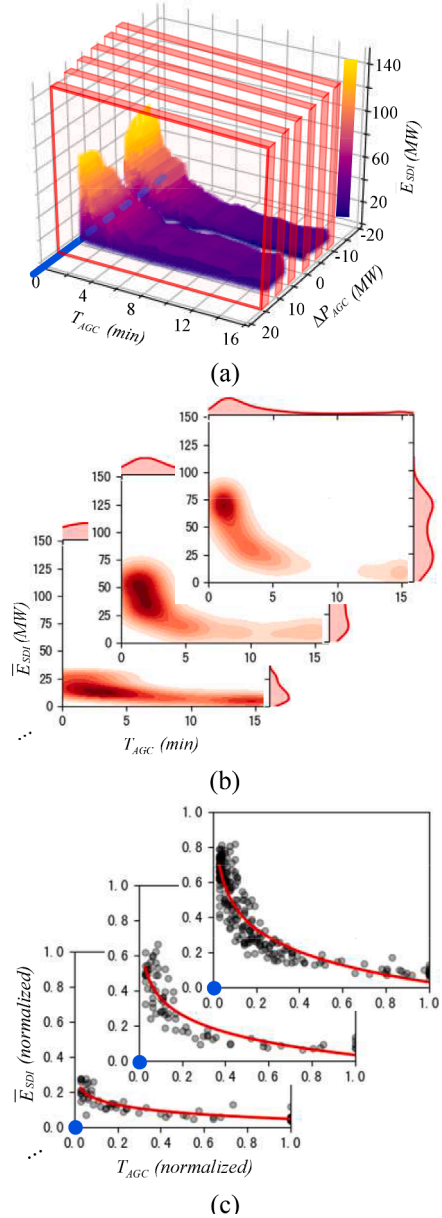


Fig. 10. Visualization of cutoff intervals in AGC sample space [84]. (a) Schematic diagram of AGC sample space segmentation; (b) Distribution density of AGC samples within different truncation intervals; (c) Fitting curves of AGC samples within different truncation intervals.

grid, providing important references for the efficiency and stability of coal-fired power plants during load peak shaving operations. Reference [106], by constructing a modified model for frequency regulation and AGC command tracking scenarios, has proposed a coordinated control approach, which is an important contribution to the assessment of AGC performance. Reference [107] introduces a strategy based on multi-variable fuzzy two-level optimization to enhance the performance of the combined operation of energy storage and thermal power generation units in grid frequency regulation. The performance and economic viability of the thermal-storage system are assessed using the rainflow counting method and the net present value approach, respectively. Reference [108] has developed a fractional-order proportional-integral-derivative controller designed to regulate hybrid power systems that include traditional thermal AGC units. The proposed controller significantly reduces the undershoot, overshoot, and settling time of the AGC system's response. Reference [109], by contrasting the performance of traditional PID controllers with that of optimized methods, exhibits the frequency regulation properties of optimized PID controllers under diverse load conditions. Similarly, reference [110] has conducted a systematic analysis of the uncertainty, load disturbances, and parameter variations in multi-area microgrid AGC, offering new methods and perspectives for AGC performance evaluation.

Based on the analysis of existing literature, it is evident that the output of AGC units in practice is continuous and nonlinear, necessitating that accurate performance evaluation closely approximates this reality. Inherent patterns exist within the vast troves of historical data, thus necessitating data-driven analysis of historical operating data of AGC units to achieve precise dynamic performance evaluation. The key to performance evaluation using data-driven methods lies in the discovery and application of these patterns to predict the behavior of AGC units under various circumstances. By employing machine learning techniques and advanced control strategies, the frequency regulation capability of the power grid can be significantly enhanced, reducing system response fluctuations, thereby improving the stability and reliability of power system operations. Simultaneously, the consideration of comprehensive economic indicators is an indispensable aspect of AGC performance evaluation, as they directly relate to the economic efficiency of the power system and the cost-effectiveness of load regulation. In summary, an accurate AGC performance evaluation requires a multifaceted methodology, which includes but is not limited to predictive analytics, simulation modeling, economic benefit evaluation, and robust control methods.

5. Conclusion and future research trends

As the proportion of wind power in regional power grids continues to increase, wind-thermal bundled power system is beneficial for balancing the fluctuations of wind power by controlling the output of thermal power units. The paper provides a detailed analysis of the operating characteristics and existing problems of wind farms and thermal power plants in frequency regulation, and points out that fluctuations in wind power affect the output of thermal power. The analysis of the paper points out that short-term power prediction of wind power can be improved in a data-driven manner through feature analysis, data cleaning, feature reconstruction, and model construction. Short term wind power prediction helps to provide advanced scheduling decisions for thermal AGC units in the power grid. The AGC system of thermal power units is a typical multi-input multi-output complex nonlinear system. Data-driven AGC performance evaluation must consider the massive historical data of the AGC unit, and the quality of the dataset must be improved. In order to construct an optimal performance evaluation method for thermal power AGC units based on multidimensional feature analysis, the performance evaluation indicators of AGC units should reflect the comprehensive tracking ability of step amplitude and duration of set point power.

The wind-thermal bundled power system still faces many technical

challenges in the future, and the future research are mainly reflected in the following aspects:

- (1) In terms of wind power prediction, especially in large-scale wind farms, due to the tight arrangement of wind turbines, the wake effect of wind turbines may lead to the influence between adjacent wind turbines, which not only reduces the energy capture efficiency of the entire wind farm, but also may lead to an increase in power prediction errors. In future research, the information of wind farms should be further refined, with the spatial and geographical location of wind turbines as one of the key features, and the temporal-spatial features of wind farms should be constructed. Combining deep learning technology with statistical empirical models to capture the characteristics of wind farm wake effects in a data-driven manner, further improving prediction accuracy.
- (2) In terms of performance evaluation of thermal AGC units, the uncertainty and variability of unit performance are still the biggest technical challenges, and factors such as fuel quality fluctuations, equipment aging, and changes in environmental conditions may all lead to differences in unit performance. When constructing AGC samples in future research work, in addition to considering AGC control instructions and actual output data, some actual engineering situations should also be taken into account. For example, the power critical point, start stop duration, operating parameters of the speed control system, and excitation system setting values of different units for starting and stopping coal mills. When constructing AGC samples in future research work, in addition to considering AGC control instructions and actual output data, some actual engineering situations should also be taken into account. For example, the start stop point and start stop duration of coal mills for different thermal AGC units, the operating parameters of the speed control system, and the setting values of the excitation system. The characteristics of AGC data should be enriched to further highlight and explore the personalized patterns of thermal AGC units.
- (3) In terms of optimal control of wind-thermal bundled power system, the essence of power system frequency control is real-time feedback regulation, so there is no standard answer to the optimal control strategy, and it must be continuously adjusted and optimized according to the needs of different scenarios. Especially considering the volatility of renewable energy, multi-objective dynamic optimization needs to be considered when allocating power between different AGC units, and reinforcement learning technology is a suitable method to solve this problem. In addition, considering the advantage of millisecond response speed in energy storage, energy storage should also be added to the control model to form a joint structure of "wind firepower storage". This helps to solve the problems of slow response speed and large errors in thermal power units, which is also the future research direction of wind-thermal bundled power system.

CRediT authorship contribution statement

Shuai Wang: Data curation, Investigation, Visualization, Writing – original draft. **Bin Li:** Conceptualization, Formal analysis, Funding acquisition, Methodology, Supervision, Writing – review & editing. **Guanzheng Li:** Data curation, Software. **Botong Li:** Project administration, Resources. **Hongbo Li:** Data curation, Project administration, Resources, Supervision. **Kui Jiao:** Formal analysis, Supervision. **Chengshan Wang:** Formal analysis, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence

the work reported in this paper.

Data availability

The authors do not have permission to share data.

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